

E-Commerce Sales Performance Report

Transforming raw transactional data into actionable business intelligence — covering customer segmentation, logistical efficiency, and sales performance.

Dataset & ETL Pipeline



Dataset Coverage

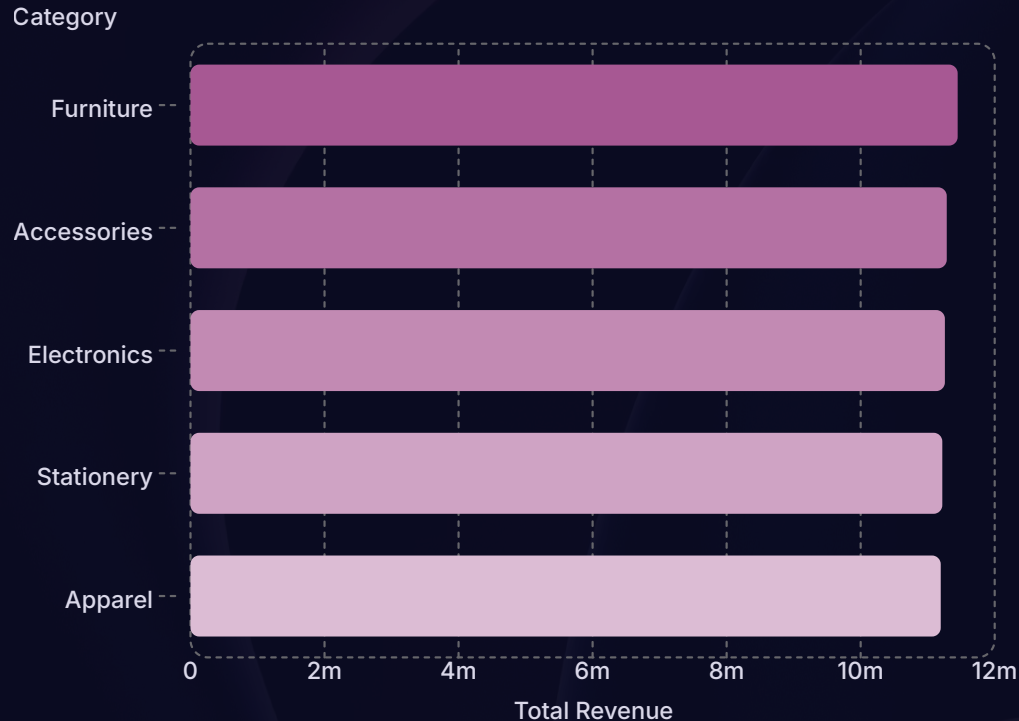
- **Customer Info:** Unique IDs and purchasing behavior
- **Sales Metrics:** Revenue, costs, and payment methods
- **Operations:** Shipment providers, priority levels, returns
- **Geography:** Country-level sales distribution

Python ETL Process

- **Cleaning:** Handled missing values; converted revenue/costs to numeric
- **Transformation:** Formatted dates; categorized transaction statuses
- **Validation:** Ensured integrity before loading into PostgreSQL

Category Profitability Analysis

Which categories generate the highest total revenue? This helps focus marketing efforts.

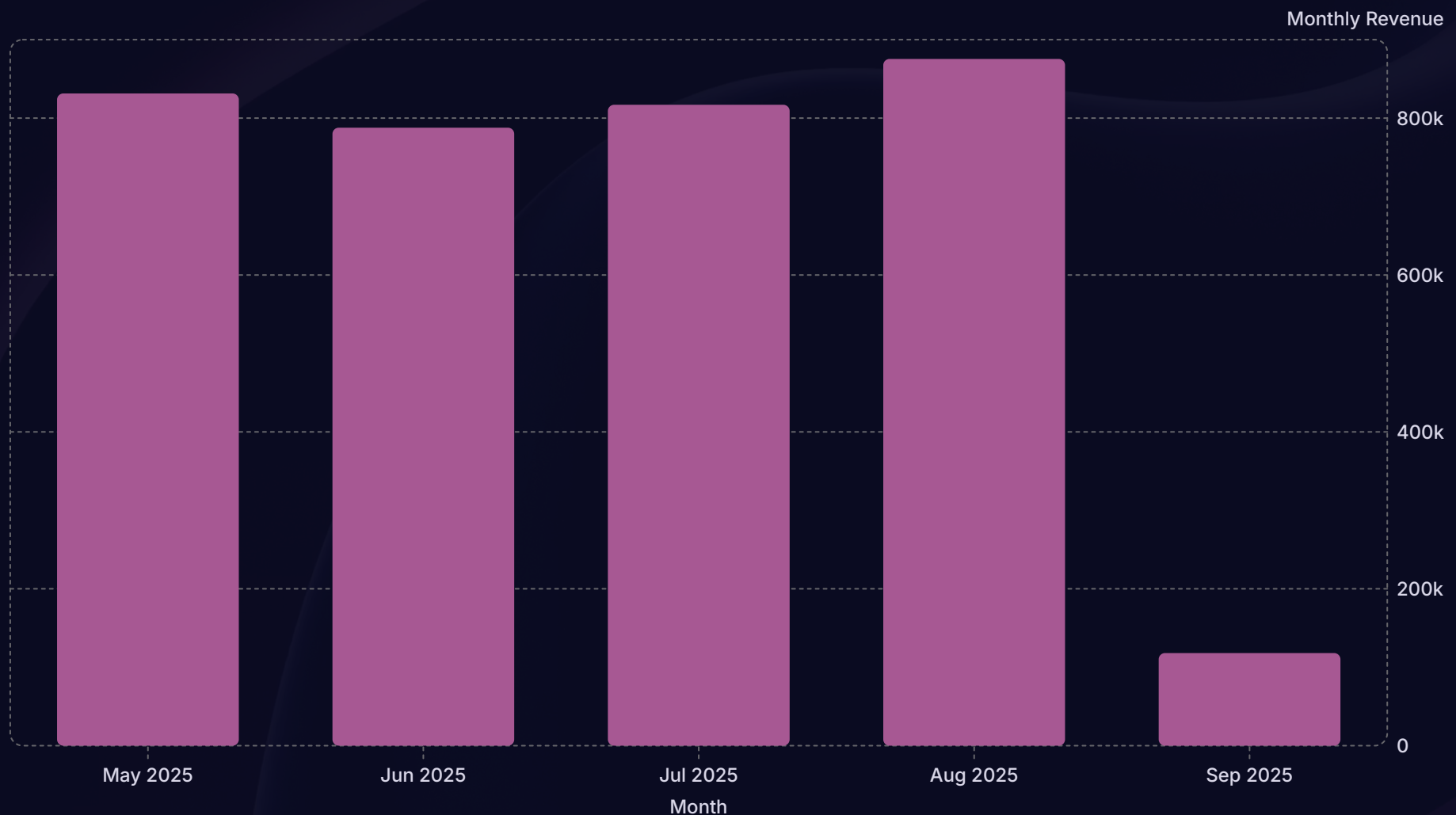


Key Insight

Revenue is remarkably balanced across all five categories — each generates between **\$11.1M and \$11.4M**, suggesting a well-diversified product portfolio with no single dependency.

Monthly Revenue Trends

What is the total revenue generated for each month of activity?



September 2025 shows significantly lower revenue (\$118K), likely reflecting incomplete month data. Revenue peaked in August at **\$875K**, with a relatively stable trend from May through August.

Customer Behavior & Segmentation

Average Ticket Size

Top customers spend ~\$4,850–\$4,890 on average, revealing a high-value user base.

High-Value Customers

Customers with **5+ purchases** and avg spend <**\$500** were identified using Window Functions. Top customer (ID 14461) averaged \$2,317 per purchase.

Why It Matters

Segmentation enables targeted retention campaigns and personalized marketing for the most profitable buyers.



Top 3 Best-Selling Products by Category

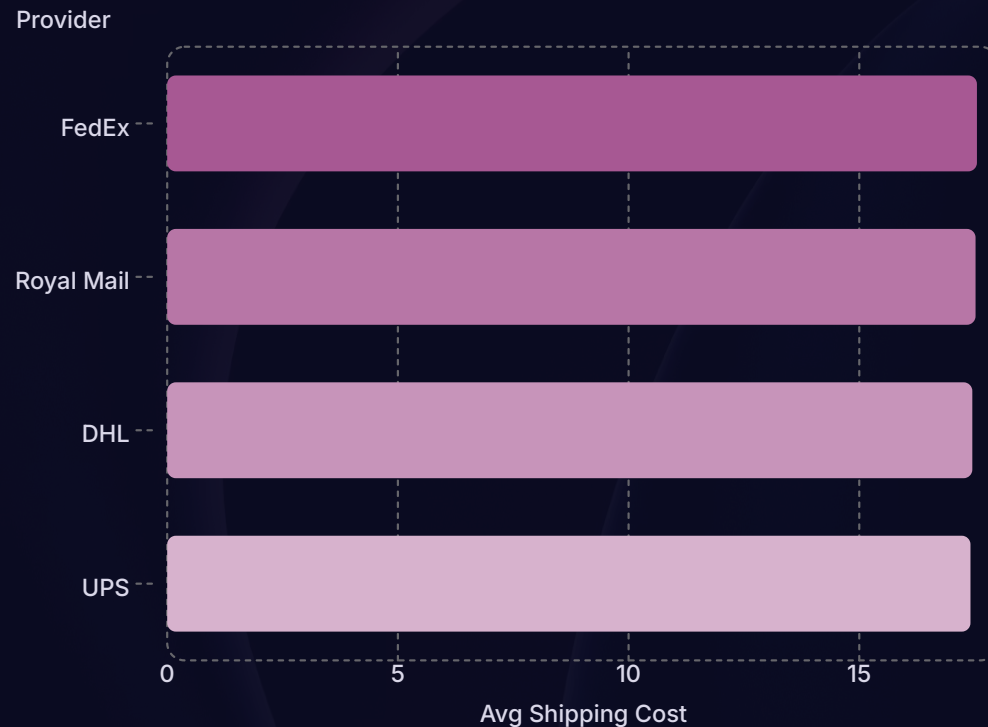
Identifies the best-selling items per category using window functions — key for inventory planning.

Category	Product	Total Orders
Accessories	Wall Clock	856
Apparel	USB Cable	848
Electronics	USB Cable	859
Furniture	Backpack	878
Stationery	White Mug	843

Each category's top seller ranges between **843–878 orders**, indicating strong and consistent demand across the catalog.

Logistics & Supply Chain

Avg Shipping Cost by Provider

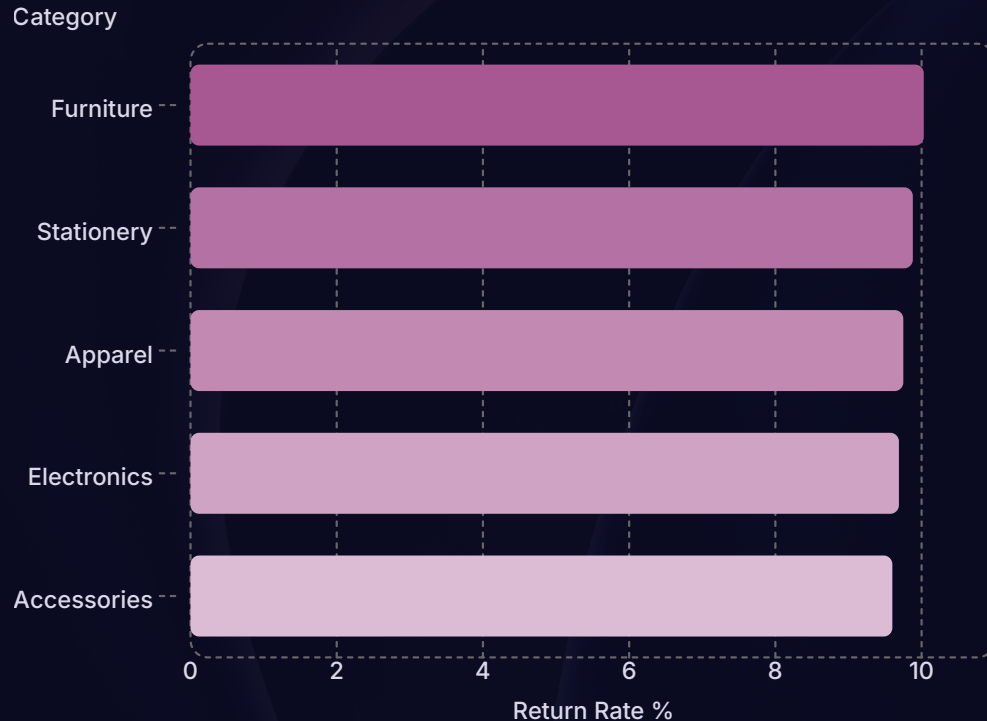


Key Findings

- **FedEx** has the highest avg shipping cost (\$17.55) but also the most completed shipments (**11,296**)
- High-priority orders cost **\$17.54** vs. \$17.37 for medium — a marginal difference
- Returned orders (\$17.46) vs. not returned (\$17.49) show **no meaningful cost gap**

Return Rates by Category

Which product category has the highest return rate? Investigating returns helps identify quality control or description issues.

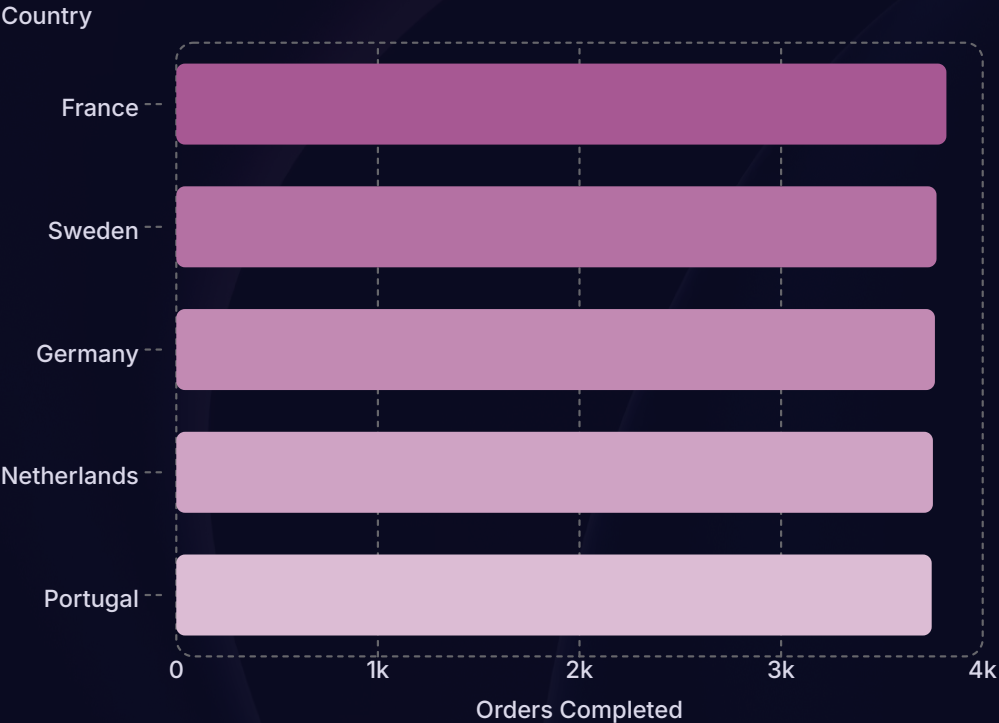


Insight

Return rates are tightly clustered between **9.6% and 10%** across all categories. **Furniture leads at 10.03%** — a potential signal to review product descriptions or packaging quality.

Geographic & Payment Insights

Top 5 Countries by Order Volume



Payment Method Performance

Method	Total Revenue
Bank Transfer	\$19.1M
PayPal	\$18.6M
Credit Card	\$18.7M

Order volume is nearly equal across top 5 European countries (~3,750 each). **Bank Transfer** leads in total revenue at \$19.1M with the highest avg spend per transaction (\$1,267).



Business Recommendations

Optimize Marketing Spend

Prioritize advertising for high-value customer segments identified through SQL analysis.

Logistical Review

Investigate providers with highest avg shipping costs (FedEx, Royal Mail) for potential savings.

Inventory Management

Increase stock for the Top 3 best-sellers per category to maximize revenue.

Retention Strategy

Focus on high-return categories (Furniture 10.03%) to address quality or description issues.